ECEN 5692: HW 4 | Matt Hartnett

Problem 1.3.A.4:

1) $x(t)$ has $r_x(3) = 4-6j$

$r_{x_I}(3) = \frac{1}{2} Re[r_x(3)]$

$= \frac{1}{2}(4)$

$\boxed{r_{x_I}(3) = 2}$

$r_{x_Q}(3) = r_{x_I}(3)$ for CSCG

$\boxed{r_{x_Q}(3) = 2}$

$r_{x_I x_Q}(3) = -\frac{1}{2} Im[r_x(3)]$

$= -\frac{1}{2}(-6)$

$\boxed{r_{x_I x_Q}(3) = 3}$

$r_{x_I x_Q}(-3) = -r_{x_I x_Q}(3)$

$\boxed{r_{x_I x_Q}(-3) = -3}$

$r_{x_Q x_I}(3) = r_{x_I x_Q}(-3)$

$\boxed{r_{x_Q x_I}(3) = -3}$

2) $r_{x_I}(-5) = 5 = \frac{1}{2} Re[r_x(-5)] \Rightarrow Re[r_x(-5)] = 10$

$r_{x_I x_Q}(-5) = -6 = -\frac{1}{2} Im[r_x(-5)] \Rightarrow Im[r_x(-5)] = 12$

$\boxed{r_x(-5) = 10 + 12j}$

3) $\tilde{r}_x(f) = 1$

$\tilde{r}_{x_I}(f) = \frac{\tilde{r}_x(f) + \tilde{r}_x(-f)}{4}$

$\cancel{= \frac{2+(-1)}{4}}$

$= \frac{1+1}{4}$

$\boxed{\tilde{r}_{x_I}(f) = \frac{1}{2}}$

Problem 1.3.A.5:

1) $\tilde{r}_{n_{wc}}(f) = \frac{1}{2} \times 10^{-21}$ W/Hz

$\tilde{r}_{n_{wc}}(f) = N_0/2 \Rightarrow N_0 = 1 \times 10^{-21}$ W/Hz

$r_{n_{wc}}(\tau) = N_0/2\, \delta(\tau) \Rightarrow r_{n_{wc}}(\tau) = \frac{1}{2} \times 10^{-21}\, \delta(\tau) \Rightarrow \boxed{c}$

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Problem 1.3.A.5 (cont.):

2) $r_{nwc}(-0.1)$

$\delta(-0.1) = \emptyset \Rightarrow \boxed{A}$

Problem 1.3.A.6:

$\tilde{r}_{nwc}(f) = N_0/2$

$E[n_{wc}(2.7)\, n_{wc}(3)] = E[n_{wc}(t)\, n_{wc}(t+\tau)]$ if $t = 2.7$, $\tau = 0.3$

$= r_{nwc}(-\tau)$

$= r_{nwc}(-0.3)$

$r_{nwc}(t) = N_0/2\ \delta(t)$

$r_{nwc}(-0.3) = \underset{=\emptyset}{N_0/2\ \delta(-0.3)}$

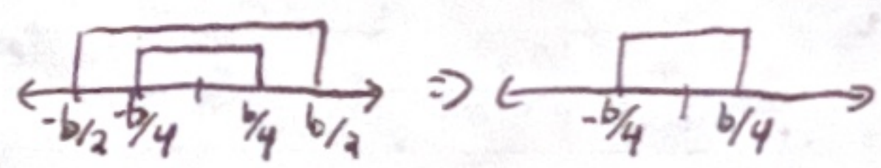
$\boxed{E(n_{wc}(2.7)\, n_{wc}(3)] = \emptyset}$

Problem 1.3.A.7:

$\tilde{r}_x(f) = 3\,\mathrm{rect}_{b/2}(f)$ $\quad$ $\tilde{h}(f) = 4\,\mathrm{rect}_b(f)$

$\tilde{r}_y(f) = \tilde{r}_x(f)\,|\tilde{h}(f)|^2$

$= 3\,\mathrm{rect}_{b/2}(f)\,|4\,\mathrm{rect}_b(f)|^2$



$= 48\,\mathrm{rect}_{b/2}(f)\,\mathrm{rect}_b(f)$

$\tilde{r}_y(f) = 48\,\mathrm{rect}_{b/2}(f) \Rightarrow \boxed{B}$

Problem 1.3.A.8:

$\tilde{h}_c(f) = 3(\mathrm{rect}_b(f-f_c) + \mathrm{rect}_b(-f-f_c))$, $\quad b = 1\,\mathrm{kHz}$, $f_c = 1\,\mathrm{MHz}$

1) $\tilde{r}_{n_c}(f) = \tilde{r}_{nwc}(f)\,|\tilde{h}_c(f)|^2$

$= N_0/2\ |3(\mathrm{rect}_b(f-f_c) + \mathrm{rect}_b(-f-f_c))|^2$

$= 9\,N_0/2\ (\mathrm{rect}_b(f-f_c) + \mathrm{rect}_b(-f-f_c))$

$\mathrm{Var}[n_c(t)] = E[n_c^2(t)]$

$= r_{n_c}(\emptyset)$

$= \mathcal{F}^{-1}\{\tilde{r}_{n_c}(f)\}(\emptyset)$

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Problem 1.3.A.8 (cont.):

$Var[n_c(t)] = F^{-1}\{\tilde{r}_{nc}(f)\}(0)$

$r_{nc}(t) = 9N_0/2\,(F^{-1}\{rect_b(f-f_c)\} + F^{-1}\{rect_b(-f-f_c)\})$

$F^{-1}\{rect_b(f-f_c)\} = b\,sinc(b\tau)e^{j2\pi f_c\tau}$

$F^{-1}\{rect_b(-f-f_c)\} = b\,sinc(b\tau)e^{-j2\pi f_c\tau}$

We only care about $r_{nc}(0)$ so we can evaluate @ $t=0$

$r_{nc}(0) = 9N_0/2\,(b(1)e^0 + b(1)e^0)$

$\boxed{Var[n_c(t)] = 9N_0 b = 9000N_0}$

2) $f_{n_c(t)}(n) = \frac{1}{\sqrt{2\pi\sigma_{n_c}^2}} e^{-\frac{|n_c - N_{n_c}|^2}{2\sigma_{n_c}^2}}$

From part 1, $\sigma_{n_c} = 9N_0 b$

Since noise is zero mean, $N_{n_c} = 0$

$f_{n_c(t)}(n) = \frac{1}{\sqrt{2\pi(81N_0^2b^2)}} e^{-\frac{n^2}{2(81N_0^2b^2)}}$

$\boxed{f_{n_c(t)}(n) = \frac{1}{9N_0b\sqrt{2\pi}} e^{\frac{-n^2}{162N_0^2b^2}}}$

3) ~~$r_{nc}(\tau) = 9N_0/2\,(b\,sinc$~~

From part 1:

$r_{nc}(\tau) = 9N_0b/2\,(sinc(b\tau))(e^{j2\pi f_c\tau} + e^{-j2\pi f_c\tau})$

$= 9N_0b/2\,(sinc(b\tau)\cos(2\pi f_c\tau))$

$E[(n_c(t_1) - E[n_c(t_1)])(n_c(t_2) - E[n_c(t_2)])]$

$= E[n_c(t_1)n_c(t_2) - n_c(t_2)E[n_c(t_1)] - n_c(t_1)E[n_c(t_2)] + E[n_c(t_1)]E[n_c(t_2)]]$

$= E[n_c(t_1)n_c(t_2)] - E[n_c(t_2)E[n_c(t_1)]] - E[n_c(t_1)E[n_c(t_2)]] + E[E[n_c(t_1)]E[n_c(t_2)]]$

$= r_{nc}(t_1 - t_2) - 0 - 0 + 0$ since $E[n_c(t)] = 0$ for zero mean Gaussian noise

$\rho = \frac{r_{nc}(t_1-t_2)}{\sqrt{9N_0b}\sqrt{9N_0b}} = \frac{r_{nc}(t_1-t_2)}{9N_0b} = \frac{(9N_0b/2)(sinc(b(t_1-t_2))\cos(2\pi f_c(t_1-t_2)))}{9N_0b}$

$\boxed{= \frac{1}{2}sinc(b(t_1-t_2))\cos(2\pi f_c(t_1-t_2))}$